BIHAR ENGINEERING UNIVERSITY, PATNA

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

B.Tech 4th Semester • Examination Resource

ANALOG ELECTRONICS & INTEGRATED CIRCUITS (2023)

Exam Date: Saturday, 31 Oct 2026 (2:00 PM - 5:00 PM)

Included Question Papers

Year	Paper Title	Course Code	Examination Session
2023	ECE Analog Circuit	`104401`	End Semester Examination - 2023 Semester-IV

Apna BEU Question Bank Archive • Compiled for B.Tech ECE 4th Semester

Contains official previous year questions with multiple-choice questions, descriptive problems, and complete circuits.

BIHAR ENGINEERING UNIVERSITY, PATNA

End Semester Examination - 2023 Semester-IV

ECE ANALOG CIRCUIT

Time: 03 Hours | Code: 104401 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Write the answer of the following (Any seven question only): **[14]**

1. Which of these are incorrect about Darlington amplifier?
   i. It has a high input resistance
   ii. The output resistance is low
   iii. It has a unity voltage gain
   iv. It is a current buffer

2. What is a cascade amplifier?
   i. A cascade of two CE amplifiers
   ii. A cascade of two CB amplifiers
   iii. A cascade of CE and CB amplifiers
   iv. A cascade of CB and CC amplifiers

3. What is the output of a class B amplifier for sinusoidal input?
   i. Sinusoidal amplifier
   ii. Half-sinusoidal
   iii. Sinusoidal with higher frequency
   iv. Square wave

4. In a class B amplifier, it is found that DC power is 25W, find the ac power.
   i. 10 W
   ii. 62.5 W
   iii. 25 W
   iv. 50 W

5. In an oscillator if phase of feedback is same as that of oscillation waveform then feedback is called
   i. Positive feedback
   ii. Negative feedback
   iii. Cannot be predicted
   iv. Either positive or negative depending upon frequency

6. For a MOSFET, the pinch-off voltage is -3V. Gate to source voltage is 20V. W/L ratio is 5. Process transconductance parameter is 40$\mu A/V^2$. Find drain to source current in saturation.
   i. 0.10 mA
   ii. 0.05mA
   iii. - 0.05mA
   iv. -50A

7. For a MOSFET, the pinch-off voltage is -3V. Gate to source voltage is 20V. W/L ratio is 5. Process transconductance parameter is 40$\mu A/V^2$. Find drain to source current in saturation.
   i. 1100 Ω
   ii. 11000 Ω
   iii. 900 Ω
   iv. 110 Ω

8. The hybrid π model is used for
   i. Analysis of BJT for low frequency
   ii. Analysis of BJT at high frequency
   iii. Analysis of BJT for feedback
   iv. Analysis of BJT for large signal

9. Which oscillator is characterized by a split capacitor in its tank circuit?
   i. RC phase shift oscillator
   ii. Colpitts oscillator
   iii. Wien bridge oscillator
   iv. none

10. The gate voltage in a JFET at which drain current becomes zero is called
    i. pinch-off voltage.
    ii. saturation voltage
    iii. active voltage.
    iv. cut-off voltage.

Q.2 Solve this question : **[14]**

1. Explain construction and working of Enhancement and Depletion type MOSFET.

Q.3 Solve this question : **[14]**

1. What is power amplifier. Explain Transformer coupled Class-A amplifier.

Q.4 Solve this question : **[14]**

1. Barkhausen criteria play an important role while designing Oscillator justify. Explain Hartley and RC phase shift Oscillator.

Q.5 Solve this question : **[14]**

1. Explain the effect of voltage series and voltage shunt feedback on gain & impedance of amplifier.

Q.6 Solve this question : **[14]**

1. The BJT is connected as a CE amplifier. Determine Z_i, Z_0, A_v, A_i hybrid model.

Q.7 Solve this question : **[14]**

1. Describe current mirror circuit in BJT. What are the condition to design current mirror circuit. Obtain the expression for current transfer ratio.

Q.8 Solve this question : **[14]**

1. Derive the expression for voltage gain and current gain and B.W of multistage amplifier.

Q.9 Solve this question : **[14]**

1. In brief explain the working of voltage series and current series circuit. Explain the same in case of BJT.